Roque Enrique López Condori

Curriculum Vitae

Personal Information

Name: Roque Enrique López Condori
Nationality: Peruvian
Website: `https://roquelopez.com`
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Education

2013–2015 **MSc in Computer Science**, *São Paulo University*, Brazil.

2006–2010 **BSc in Computer Science**, *San Agustin National University*, Peru.

Awards

- 2023, Distinguished Participant, External Operations Platform Data Challenge (JIATF), USA.
- 2023, Best Paper Honorable Mention Award, IEEE Visualization Conference (VIS), Australia.
- 2023, Best Paper Award, Data Management for End-to-End Machine Learning (DEEM) Workshop, co-located with SIGMOD, USA.
- 2020, Member of finalist team, Wells Fargo Campus Analytics Challenge, USA.
- 2016, Finalist, Best MSc Dissertation Award, IEEE Latin American Conference on Computational Intelligence, Colombia.
- 2014, Young Scientist Best Paper Award, VII International Conference on Intelligent Information and Engineering Systems, Poland.
- 2014, Samsung Brazil Master's Scholarship (2014–2015), Brazil.
- 2013, Top 5 Bachelor's Thesis, Simposio Peruano de Inteligencia Artificial, Peru.
- 2010, Graduated in the top 5 of the undergraduate program, Peru.
- 2005, First Place and Scholarship, High School Graduation, Peru.

Experience

Dec 2018 – Present **Research Engineer**, *New York University*, USA.
Conduct research and software engineering at NYU's Visualization, Imaging, and Data Analytics (VIDA) Lab on multiple DARPA and ARPA-H funded projects.
- Core contributor to BDI-Kit, an open-source Python toolkit for biomedical data integration and harmonization using Large Language Models (LLMs) and Natural Language Processing (NLP).
- Extended AlphaD3, an automated machine learning (AutoML) system, by adding support for multiple machine learning tasks, enabling automated model selection and end-to-end pipeline construction.
- Investigated and implemented LLM-based reasoning methods for procedural text understanding, including error detection and extraction of action preconditions and effects.

Feb 2017 – Aug 2018 **Research Engineer**, *Inria*, France.
Conducted research on extracting symbolic knowledge from unstructured text using natural language processing, machine learning, machine reading, ontologies, and Semantic Web technologies.

Oct 2016 – Sept 2018 **CTO**, *DoctorCV*.
Co-led the technical development of a startup that uses NLP techniques to evaluate résumés and recommend improvements to increase interview success rates.

Jul 2015 – Jan 2017 **Researcher, Software Engineer**, *Elabora Consultoria*, Brazil.
Researched information retrieval methods and word embedding techniques to automatically construct patent term hierarchies. Developed scalable solutions using Python, Apache Spark, and HBase.

Feb 2014 – Jul 2014 **Teaching Assistant**, *University of São Paulo*, Brazil.
Teaching assistant for the course *Professional Information in Computer Science*, taught by Prof. Thiago Pardo. Assisted with grading assignments, leading discussion sessions, holding office hours, and supporting approximately 50 students.

Oct 2013 – Dec 2013 **Software Engineer**, *Dicionário Criativo*, Brazil.
Developed a Python-based Twitter crawler to collect public discussions about books, music, and films. Built text normalization modules, including spell correction, to improve data quality.

Jan 2012 – Dec 2012 **Researcher, Software Engineer**, *Lindexa – Wayra Telefónica*, Peru.
Researched and developed opinion mining and sentiment analysis techniques for Spanish. Implemented Python crawlers for Facebook and Twitter to collect large-scale public social media data. Lindexa was one of the winning startups in the first Wayra Peru accelerator program.

Jan 2011 – Dec 2011 **Technical Researcher, Developer**, *Cátedra CONCYTEC*, Peru.
Conducted research on multi-document extractive summarization, text classification, and document clustering. Developed algorithms for selecting representative and informative content from large text collections.

Research Projects

Mar 2024 – Present ***BDF: Biomedical Data Fabric Toolbox***, *New York University*, USA.
ARPA-H-funded project developing methods and tools to streamline the integration and harmonization of biomedical research data from thousands of heterogeneous sources.

Dec 2021 – Jun 2024 ***PTG: Perceptually-enabled Task Guidance***, *New York University*, USA.
DARPA-funded project focused on developing AI assistants capable of providing just-in-time visual and audio guidance to support task execution.

Dec 2018 – Mar 2024 ***D3M: Data-Driven Discovery of Models***, *New York University*, USA.
DARPA-funded project focused on developing automated machine learning systems that enable domain experts to build predictive models without requiring data science expertise.

Feb 2017 – Aug 2018 ***ALOOF: Autonomous Learning of the Meaning of Objects***, *Inria*, France.
Research project focused on enabling robots and autonomous systems to learn object semantics from unstructured text, ontologies, and Semantic Web resources.

Jul 2015 – Feb 2017 ***Patent Faceted Search Using Topic Modeling***, *Elabora Consultoria*, Brazil.
Research project focused on developing information retrieval methods for faceted search and exploration of large-scale patent collections.

Jan 2014 – Jun 2015 ***Semantic Text Processing in Brazilian Portuguese***, *Samsung – ICMC/USP*, Brazil.
Research project focused on advancing semantic processing of Brazilian Portuguese, with applications to opinion mining and automatic text summarization.

Jan 2011 – Dec 2011 ***Automatic Summarization of Medical Records Using Natural Language Processing Techniques***, *Cátedra CONCYTEC*, Peru.
Research project focused on investigating automatic multi-document summarization methods for medical records using natural language processing.

Monographs

MSc Thesis *Title:* Automatic Aspect-Based Opinion Summarization Methods
Description: This master's thesis investigates extractive and abstractive opinion summarization using an aspect-based approach. In addition to applying existing methods, two new approaches for Portuguese were proposed, achieving the best performance in the experiments.

BSc Thesis *Title:* Medical Document Classification Based on Keywords Using Semantic Information
Description: This thesis presents a method for classifying medical documents that improves over existing approaches. The method combines statistical information with semantic relatedness between document keywords.

Languages

Spanish	Fluent	*native language*
English	Advanced Level	*completed studies*
Portuguese	Advanced Level	*completed studies*

Programming Knowledge

ML	PyTorch, Scikit-learn, Numpy, Pandas
GenAI	LangChain, MCP, LiteLLM
NLP	Transformers, NLTK, SpaCy
RL	Gymnasium, Ray
Languages	Python, Java, C++
Others	Git, Docker, Linux

Main Interests

- Data Integration
- Data-centric AI
- Applied Machine Learning
- Natural Language Processing
- Reinforcement Learning
- Artificial Intelligence